

Student A - Assignment Submission

Q1. What does the coefficient of determination (R^2) measure, and what do values close to 1 versus close to 0 (or negative) tell you about the model?

Answer: R^2 , the coefficient of determination, measures the amount of variance in the prediction that is explained by the dataset (computed from the residuals - the differences between samples and predictions). A value close to 1 means an almost perfect regression, while a value close to 0 - or negative - implies a bad model.

Q2. Compare Ridge, Lasso, and ElasticNet regularization. For each, state which norm of the weights it penalizes and its characteristic effect on the weights.

Answer: Ridge adds a penalty on the squared L2 norm of the weights; it shrinks the weights and prevents them from exploding due to multicollinearity or ill-conditioning. Lasso penalizes the L1 norm, which promotes sparsity - it drives some weights to exactly zero, so the model selects features. ElasticNet combines both L1 and L2 penalties (via α and l1_ratio), so it is sparse like Lasso but keeps Ridge's regularization and avoids selectively excluding correlated features.

Q3. What problem does RANSAC address in linear regression, and how does it work at a high level?

Answer: RANSAC (random sample consensus) deals with outliers, which bias an ordinary least squares fit. It iteratively splits the data into inliers and outliers, trains only on the inlier samples, then re-evaluates all samples to see which are still inliers. It repeats until a fixed number of iterations or a desired score is reached, giving a fit that ignores outliers.

Q4. Why is logistic regression considered a classification method despite its name, and what is the role of the sigmoid (logistic) function?

Answer: Although called regression, it classifies by estimating the probability that a sample belongs to a class. Probabilities must lie in $(0,1)$, so the linear term z is passed through the sigmoid (logistic) function, which maps $\mathbb{R}$ into $(0,1)$: 0.5 at $z=0$ with asymptotes at 0 and 1. Thresholding the probability (e.g. at 0.5) gives the class.

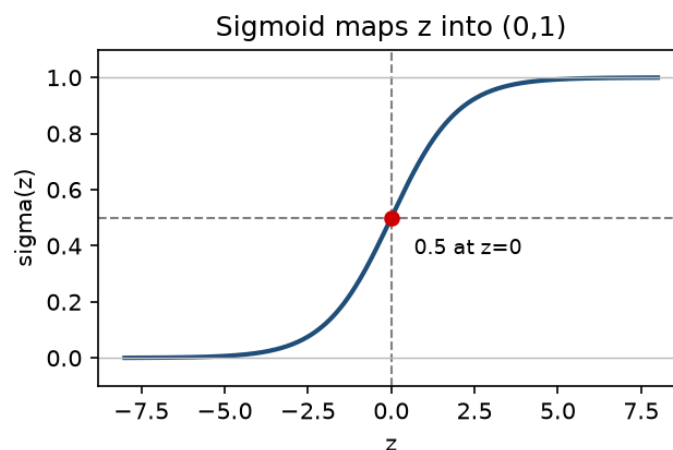


Figure 1: the sigmoid squashes z into $(0,1)$; 0.5 at $z=0$, asymptotes at 0 and 1.

Q5. In scikit-learn's LogisticRegression, what does the parameter C represent, and how do larger versus smaller values change the model?

Answer: C is the inverse regularization factor ($1/\alpha$). Larger C reduces the regularization strength (weights can grow); smaller C - especially below 1 - increases regularization and forces the weights closer to the

origin. The penalty can be L1 or L2.